

Credit Card Approval

Predicting Delinquency with Logistic Regression

Applied Business Statistics, Group Assignment 2

INTRODUCTION

Credit card risk management is a core task for lenders; it is thus fundamental to identify the possibility of applicants to fall behind on payments because it can materially affect portfolio losses. This report studies the demographics that, once submitted at application, are used to discriminate between clients who may or may not enter serious delinquency. Since the outcome of the research is binary (either delinquent or not) we can estimate a logistic regression model, using the approach studied in the course.

RESEARCH QUESTION

The research question on which we developed our research is: "To what extent can demographic and socio-economic variables collected during the application stage can predict whether a credit card applicant will subsequently become delinquent customer (60+ days past due)?"

This question is relevant for two main reasons. From the statistical point of view, the target is very imbalanced that gives us a natural setting to discuss the ROC curve, the AUC and the choice of the classification threshold. From an economic perspective, lenders are interested in simple application-level signals because they can be scored before any credit history is available with the bank.

DATASET

Source: Credit Card Approval Prediction, rikdifos, Kaggle. <https://www.kaggle.com/datasets/rikdifos/credit-card-approval-prediction/data>

The dataset comes in two files. The application record contains 438,557 applicants with 18 fields (gender, car/realty ownership, children, annual income, income type, education, family status, housing type, age in days, days employed, etc.). The credit record contains 1,048,575 monthly observations stating the delinquency status of each applicant. The binary target is constructed per applicant: `bad_client` = 1 if the monthly status ever reached 2, 3, 4 or 5 (which codes 60-89, 90-119, 120-149 and 150+ days past due respectively) and 0 otherwise. After merging the two files on applicant ID and dropping duplicate applications, the working sample contains 36,457 applicants of whom 617 (1.69%) are classified as bad.

DATA PREPARATION

Only the `OCCUPATION_TYPE` variable had missing values (about 31% of the sample), since dropping the values would have modified the dataset too heavily we decided to keep the variable but not using it as a regressor. `DAYS_BIRTH` was converted to `AGE_YEARS` (in years). `DAYS_EMPLOYED` has negative values for employed individuals and took a large positive sentinel value (365,243) for pensioners and unemployed, so we created a dummy `IS_EMPLOYED` and, for employed individuals we created the variable `YEARS_EMPLOYED`. Income was rescaled to thousands (`INCOME_K`) to keep the coefficient on a readable scale, as in the lecture examples. `CODE_GENDER`, `FLAG_OWN_CAR` and `FLAG_OWN_REALTY` were encoded as binary dummy variables (0/1). `HIGH_EDU` = 1 for Higher education or Academic degree, `MARRIED` = 1 for Married or Civil marriage. The correlation matrix showed `CNT_CHILDREN` and `CNT_FAM_MEMBERS` are almost collinear ($r \sim 0.88$); we decided to only keep `CNT_FAM_MEMBERS`, which describes the number of children. `FLAG_MOBIL` was dropped due to it being a constant.

Descriptive statistics on the continuous predictors (N = 36,457):

Variable	Mean	Std	Min	Median	Max
AGE_YEARS	43.77	11.51	20.50	42.60	68.90
YEARS_EMPLOYED	6.03	6.48	0.00	4.30	43.00
INCOME_K (\$ thousand)	186.69	101.79	27.00	157.50	1,575.00
CNT_CHILDREN	0.43	0.74	0	0	19
CNT_FAM_MEMBERS	2.20	0.91	1	2	20

Table 1. Descriptive statistics of the continuous predictors.

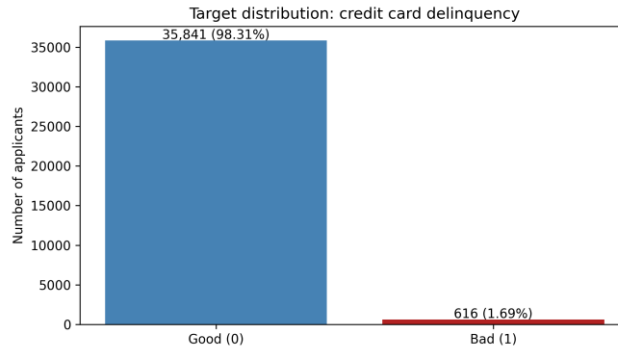


Figure 1. Target distribution. The sample is strongly imbalanced (1.69% bad clients).

METHODOLOGY

The dataset is split while preserving the class proportions into training (60%), validation (20%) and test (20%) sets. The logistic regression used in our research is estimated on the training set by maximum likelihood using statsmodels. The other two sets are used to tune the probability of threshold (validation set), and to produce an unbiased estimate of generalisation performance at the end (test set). The model specification used in our report is:

$$\text{logit } P(\text{bad_client}=1) = \beta_0 + \beta_1 \text{ AGE} + \beta_2 \text{ YEARS_EMP} + \beta_3 \text{ INCOME_K} + \beta_4 \text{ FAM_MEMBERS} + \beta_5 \text{ GENDER_M} + \beta_6 \text{ OWN_CAR} + \beta_7 \text{ OWN_REALTY} + \beta_8 \text{ HIGH_EDU} + \beta_9 \text{ MARRIED} + \beta_{10} \text{ IS_EMPLOYED}$$

REGRESSION RESULTS

The table below reports coefficients, standard errors, z-statistics, p-values and odds ratios, except of the coefficient. The overall model is significant ($p = 2.07\text{e-}08$) and the McFadden pseudo-R-squared is 0.0149, which is modest but expected due to the fact that we only observe pre-contractual demographic, (meaning acquired during the application process) information and not behavioural variables (previous delinquencies, utilisation, etc.).

Variable	Coef	Std.Err	z	p-value	Odds ratio
const	-3.8967	0.4168	-9.349	0.0000	0.020
AGE_YEARS	0.0013	0.0062	0.202	0.8399	1.001
YEARS_EMPLOYED	-0.0504	0.0120	-4.203	0.0000	0.951
INCOME_K	0.0002	0.0005	0.308	0.7578	1.000
CNT_FAM_MEMBERS	0.1360	0.0582	2.338	0.0194	1.146
GENDER_M	0.2691	0.1198	2.247	0.0246	1.309
OWN_CAR	-0.0942	0.1190	-0.792	0.4286	0.910
OWN_REALTY	-0.4436	0.1076	-4.122	0.0000	0.642
HIGH_EDU	-0.0675	0.1240	-0.545	0.5859	0.935
MARRIED	-0.3968	0.1402	-2.831	0.0046	0.672
IS_EMPLOYED	0.2904	0.2122	1.368	0.1712	1.337

Table 2. Logistic regression estimates on the training set ($N = 21,873$).

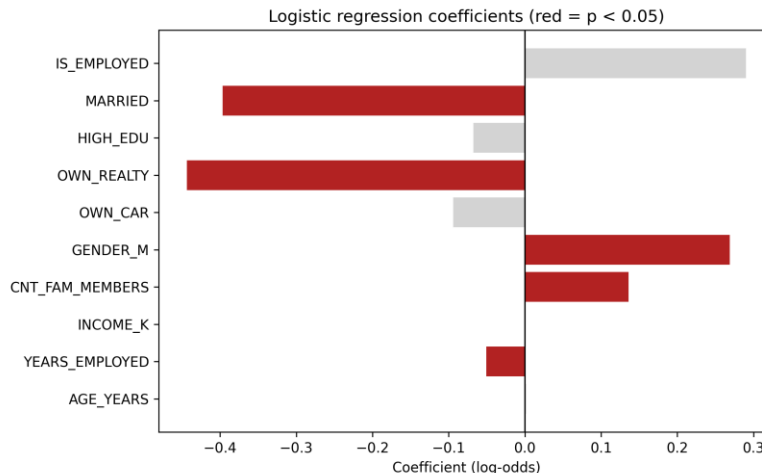


Figure 2. Estimated log-odds coefficients (red bars are significant at $p < 0.05$).

PREDICTIVE PERFORMANCE

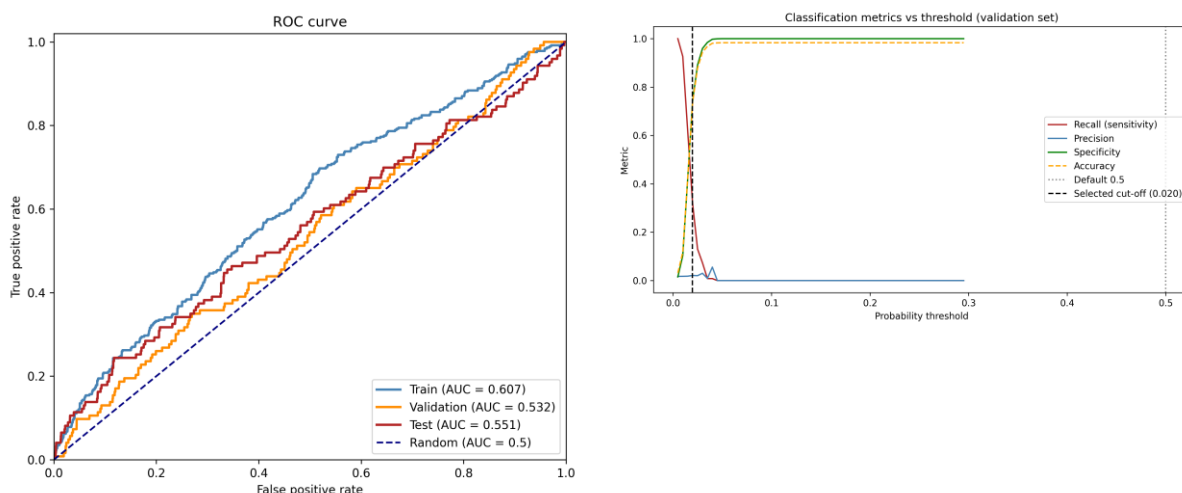


Figure 3 (left): ROC curves for training (AUC = 0.607), validation (0.532) and test (0.551). Figure 4 (right): classification metrics on the validation set as a function of the probability threshold; the threshold that maximises sensitivity + specificity is 0.020.

The AUC gap between the training set (0.607) and validation/test sets (0.53 and 0.55) shows mild overfitting instead of a useless model; on the test set the probability of ranking a random bad client over a random good one is 55%, which is better than chance but low in absolute terms. The 0.5 default threshold classifies every applicant as a good one and therefore has zero recall. It does not include all 123 true bad clients in the test set. Using the validation set, the threshold that maximises sensitivity + specificity is 0.020, just above the sample base rate. Applying it to the test set recovers 34.96% of true bad clients at the cost of flagging 27% of good ones as suspicious. A compromise that a risk officer would consider acceptable, seeing how credit bureaus usually operate upstream of the lender's own model.

	threshold = 0.5	threshold = 0.020
Accuracy	0.9831	0.7246
Precision	0.0000	0.0218
Recall (sensitivity)	0.0000	0.3496
Specificity	1.0000	0.7311
Bad clients caught (out of 123)	0	43

Table 3. Test-set performance at default and tuned thresholds.

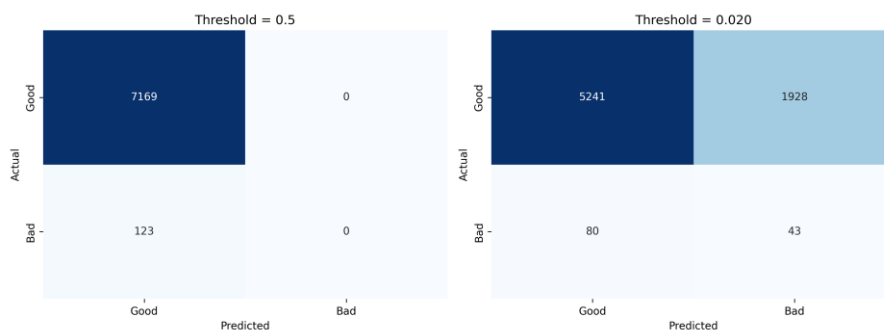


Figure 5. Confusion matrices on the test set. Left: default 0.5 cutoff, no bad client is recovered. Right: tuned cutoff (0.020), 43 of 123 bad clients are flagged.

INTERPRETATION AND CONCLUSION

Four of the studied coefficients are relevant at the 5% level and have a clear economic meaning. Each additional year of employment decreases the odds of serious delinquency by about 4.9%, consistent with job stability being a good indicator for income continuity. Owning real estate reduces the likelihood by almost 36%, since home-owners have stronger financial stability making it less probable for them to fall behind on payments. Being married also lowers the probability by 33%, in line with household smoothing literature. Larger families increase the odds, suggesting evidence of financial pressure overweighting the smoothing effect of having an increased number of potential earners. Gender plays a small role, stating that male applicants face mildly higher odds compared to female applicants. Age, income, car ownership, higher education and the employment status are not statistically different from zero once the other variables are controlled.

Overall we can state that demographic data collected during the application carry little power when predicting credit card delinquency; however the model produces economically relevant results rendering it useful. Its practical usefulness is based entirely on moving away from the default 0.5 cutoff. At the validation threshold of 0.02 the model cuts off 27% of applicants as risky and catches 35% of true delinquencies. Such statistics will be used as a starting point and be combined downstream with credit scores and behavioural variable collected post application.

BIBLIOGRAPHY

Dataset link: <https://www.kaggle.com/datasets/rikdijos/credit-card-approval-prediction/data>